MATH211
Discrete Mathematics

Functions & Sequences

Agenda

2.3 Functions

2.4 Sequences and Summations

Discrete Mathematics and Its Applications

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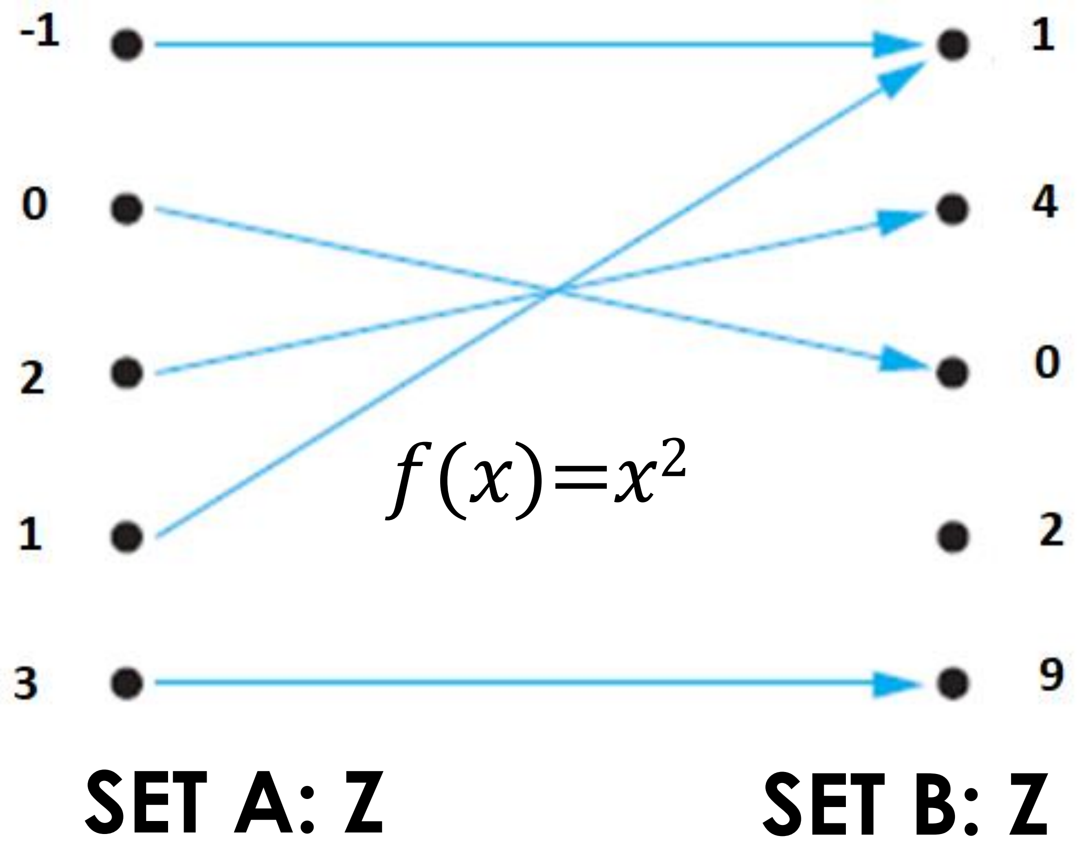
2.3. Functions

- One-to-One
- Onto
- Inverse
- Composition
- Ceil & Floor

Definition of Functions

- Let A and B be non-empty sets. A function ***f*** from A to B is an assignment of exactly one element of B to each element of A **[Each element of the Set A has exactly one value from the Set B].**
  - We write $f(a)=b$ if b is the unique element of the set B assigned by the function f to the element a of the set A.
    - **b** is the image of **a** and **a** is the pre-image of **b**.
  - If f is a function from A to B, we write $f:A \rightarrow B$.f : A → B (***f*** maps A to B)
  - A: Domain of ***f***
  - B: Codomain of ***f***
  - Range of ***f***: Set of all images of elements of A.
- Functions are sometimes called mappings or transformations

Definition of Functions

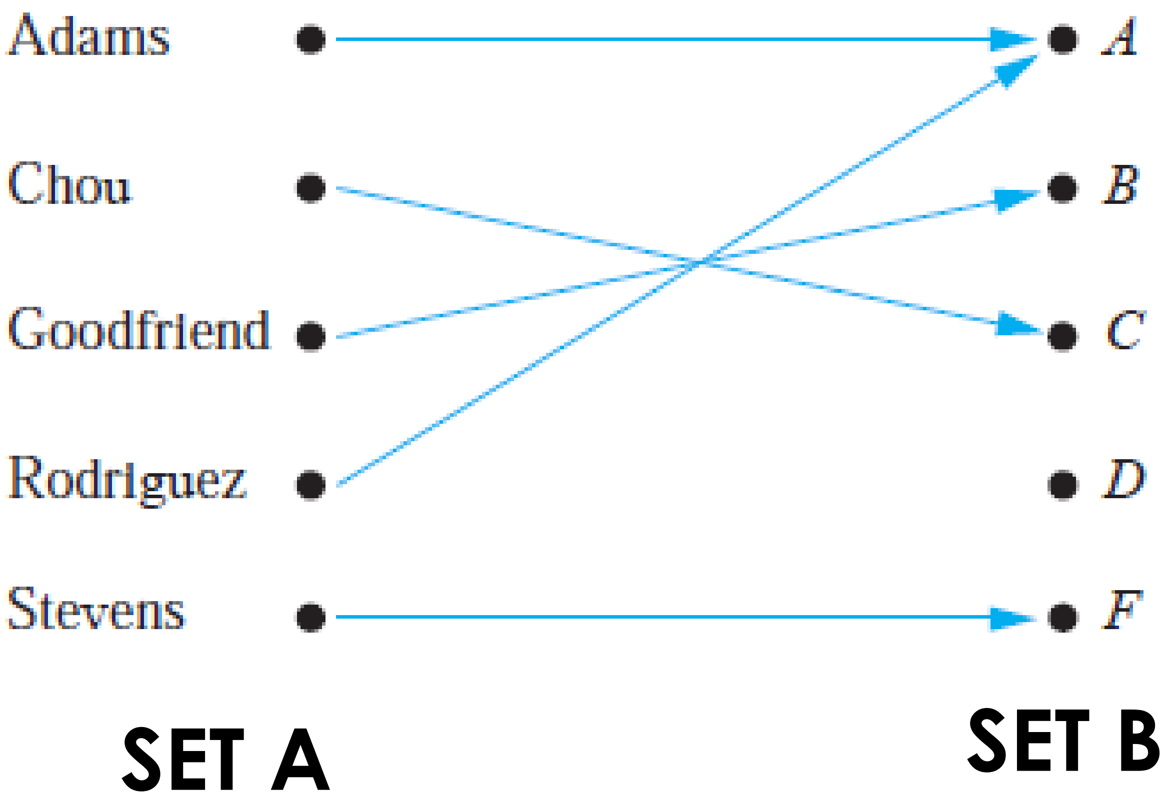


Let f be a function from **Z** to **Z** such that $f(x)=x^2$

Each element of the Set A has exactly one value from the Set B.

f: A → B (*f* maps A to B)
f(a) = b

Definition of Functions

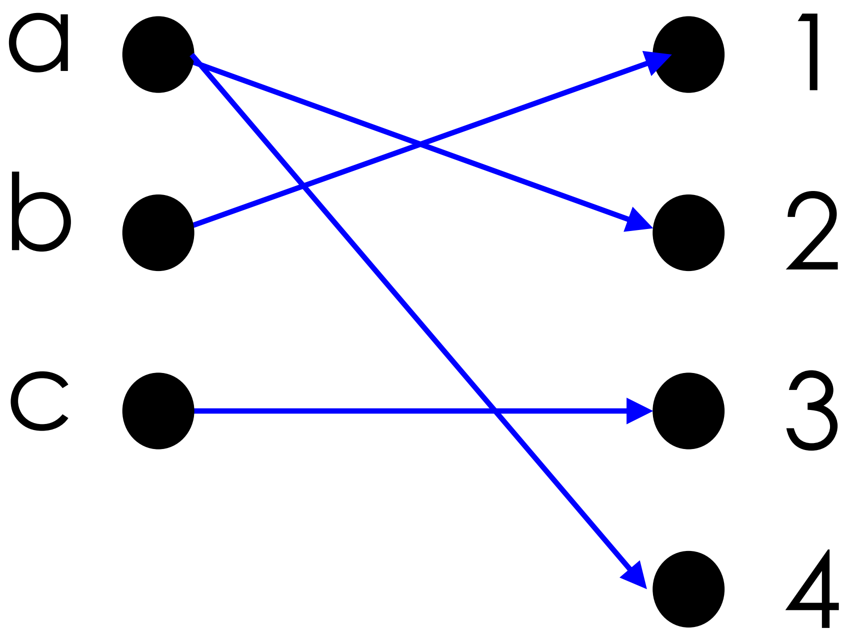


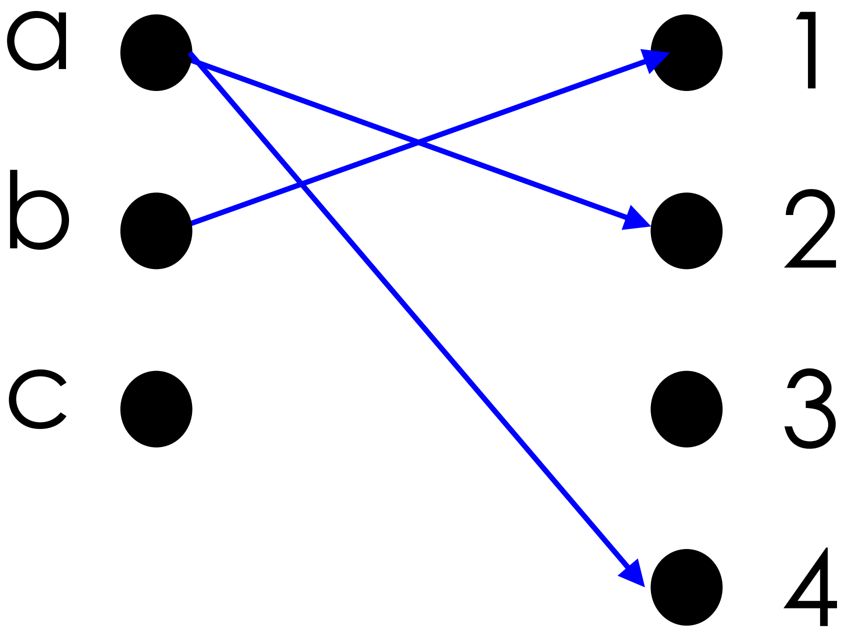
Let A and B be non-empty sets. A function ***f*** from A to B is an assignment of exactly one element of B to each element of A.

f: A → B (*f* maps A to B)
f(a) = b

Assignment of Grades in Discrete Math Course

Definition of Functions

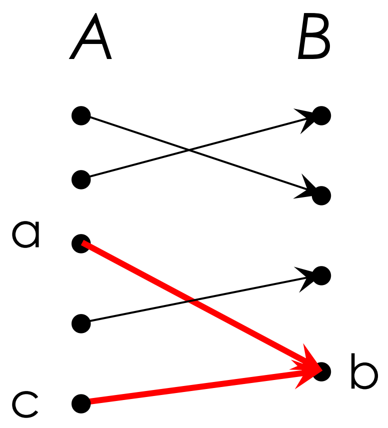




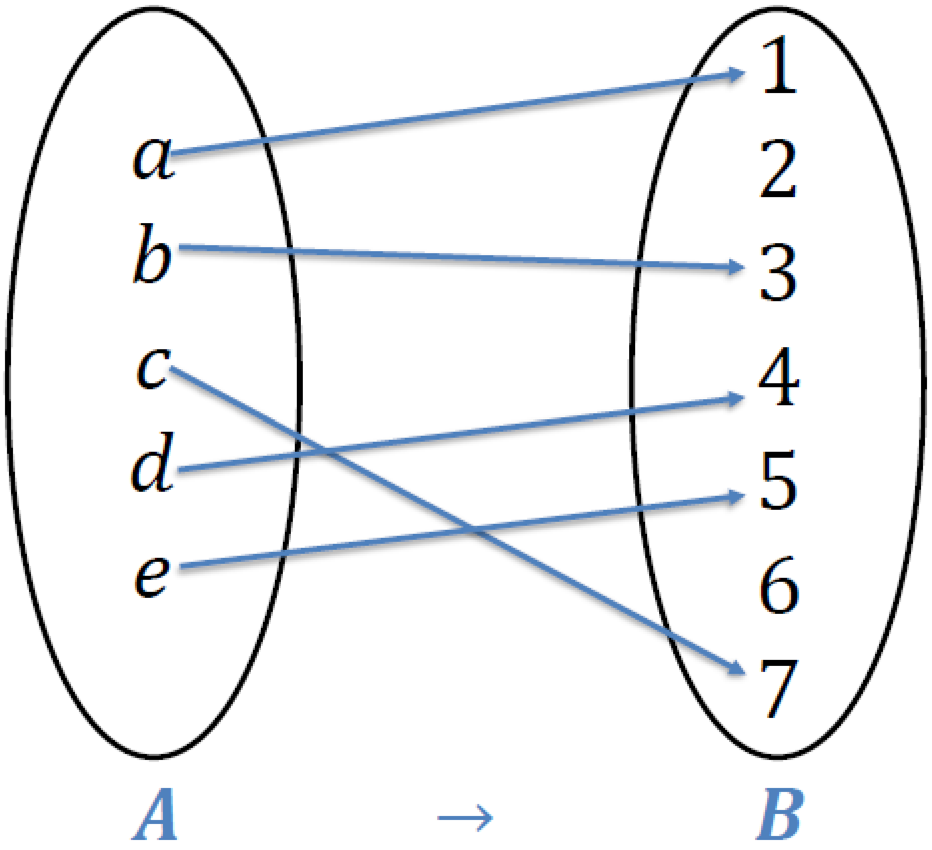
Not valid functions

Function Terminology

- If f:$A \rightarrow B$, and $f(a)=b$ (where $a \in A$ & $b \in B$), then:
  - A is the *domain* of f
  - B is the *codomain* of f
  - b is the *image* of a under f
  - a is a *pre-image* of b under f
    - In general, **b may have more than one pre-image**
    - In general, **a can not have more than one image**
  - *Range of f:* Set of all images of elements in A.



Function Terminology



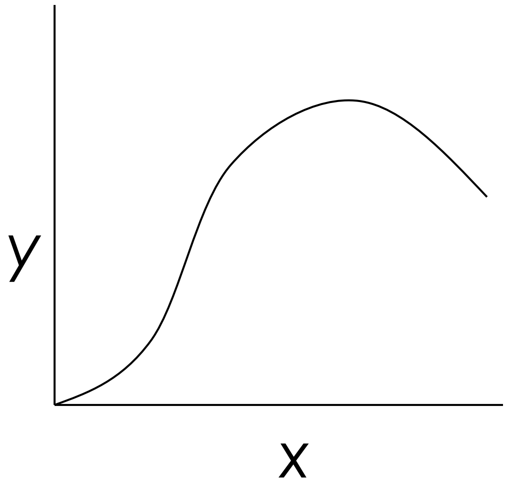
$\text{Domain} = \{a, b, c, d, e\}$

$\text{Co-Domain} = \{1,2,3,4,5,6,7\}$

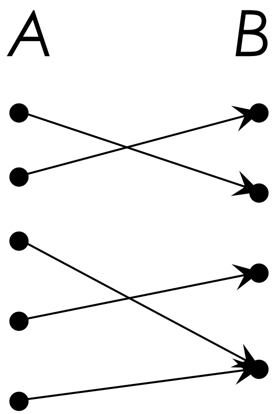
$\text{Range} = \{1,3,4,5,7\}$

Graphical Representations

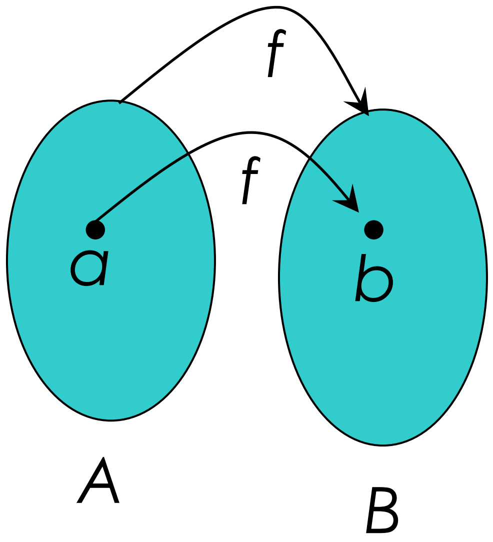
- Functions can be represented graphically in several ways:



Plot

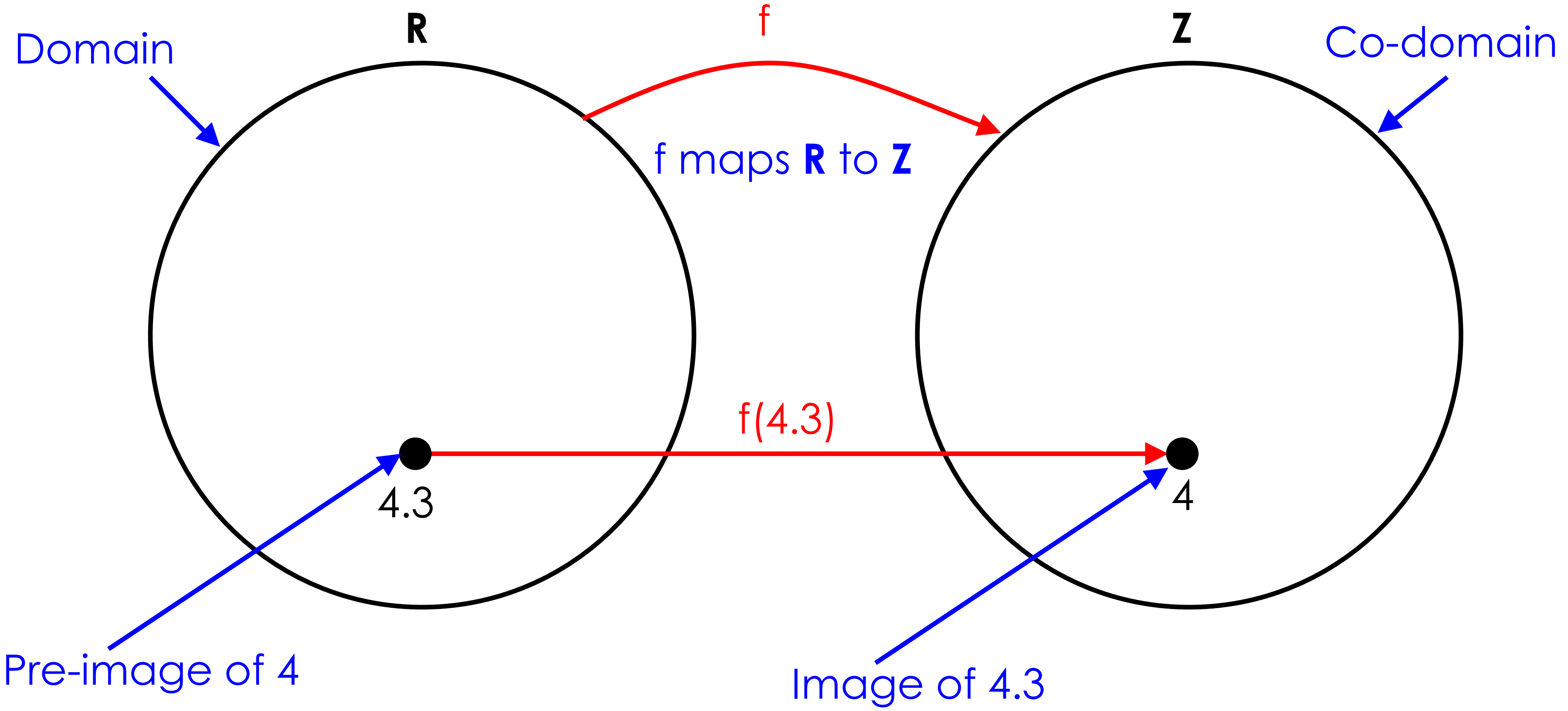


Graph



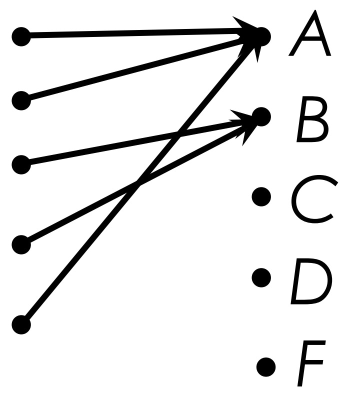
Venn diagrams

Function terminology

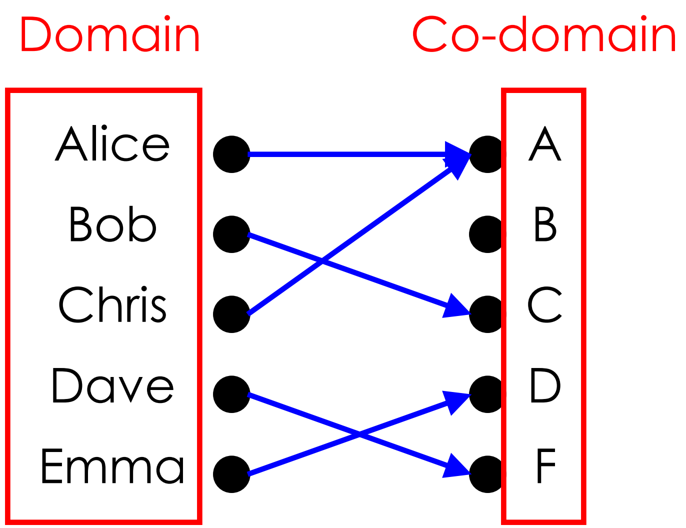


Range vs. Codomain - Example

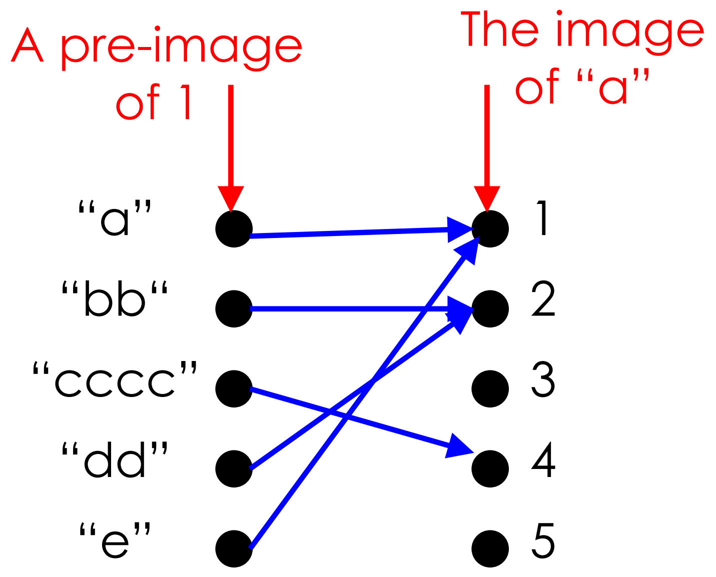
- Suppose that: "*f is a function mapping students in this class to the set of grades {A,B,C,D,E}.*"
- At this point, you know *f*'s codomain is: {A,B,C,D,F}, and its range is unknown!.
- Suppose the grades turn out all As and Bs.
- Then the range of *f* is {A,B}, but its codomain is still {A,B,C,D,F}!.



More functions

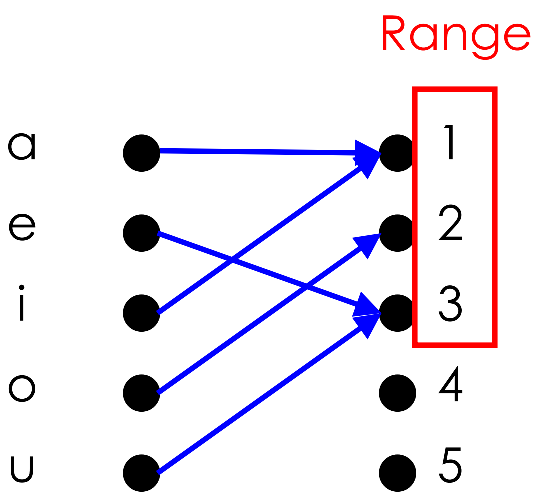


A class grade function

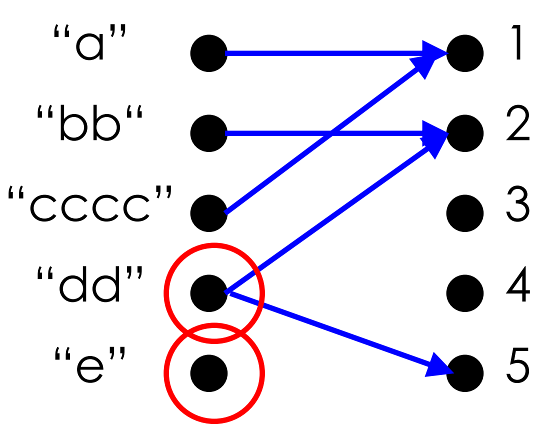


A string length function

More functions



Some function…



Not a valid function!
Also not a valid function!

Function Arithmetic

- Let f_1 and f_2 be functions from **R** to **R**. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from **R** to **R** defined for all $x \in$ **R** by
- $(f_1 + f_2)(x) = f_1(x) + f_2(x)$,
- $(f_1 f_2)(x) = f_1(x)\, f_2(x)$.

Function Arithmetic

- Let $f_1(x) = 2x$
- Let $f_2(x) = x^2$

- $f_1 + f_2 = (f_1+f_2)(x) = f_1(x)+f_2(x) = 2x+x^2$

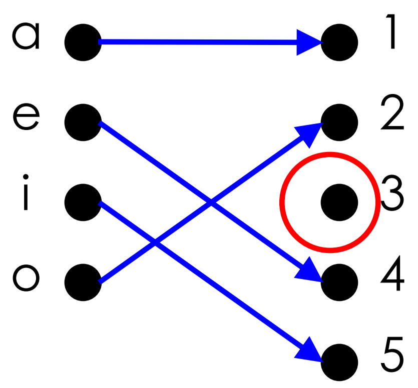
- $f_1 * f_2 = (f_1*f_2)(x) = f_1(x)*f_2(x) = 2x*x^2 = 2x^3$

Some Functional Properties

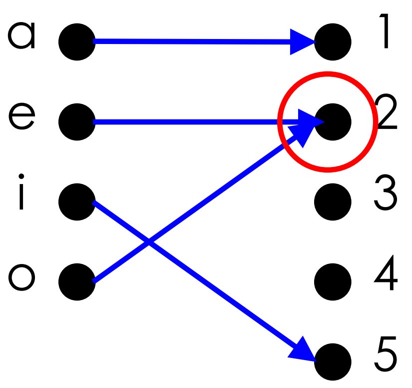
- **one-to-one** or **injective**
- **onto** or **surjective**
- **one-to-one correspondence** or **bijection**

One-to-one functions

- A function f is said to be one-to-one or injective if and only if f(x)=f(y) implies that x=y for all x and y in the domain of f. The function is said to be an injection.
- A function is one-to-one or injection *if and only if* all the elements [images] in the co-domain have a unique pre-image
- Note that there can be un-used elements in the co-domain

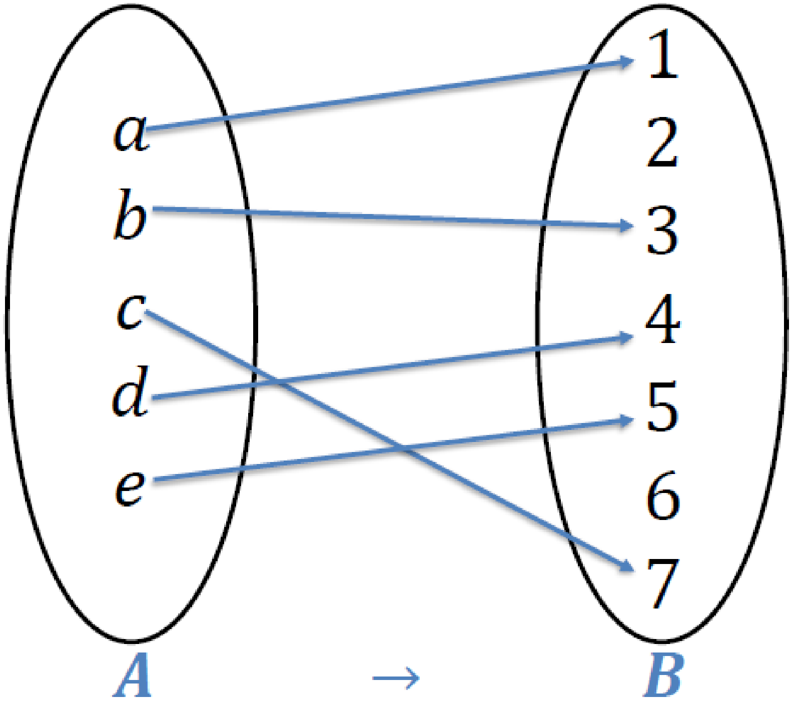


A one-to-one function



A function that is not one-to-one

More on One-to-One



$f(a) = 1$

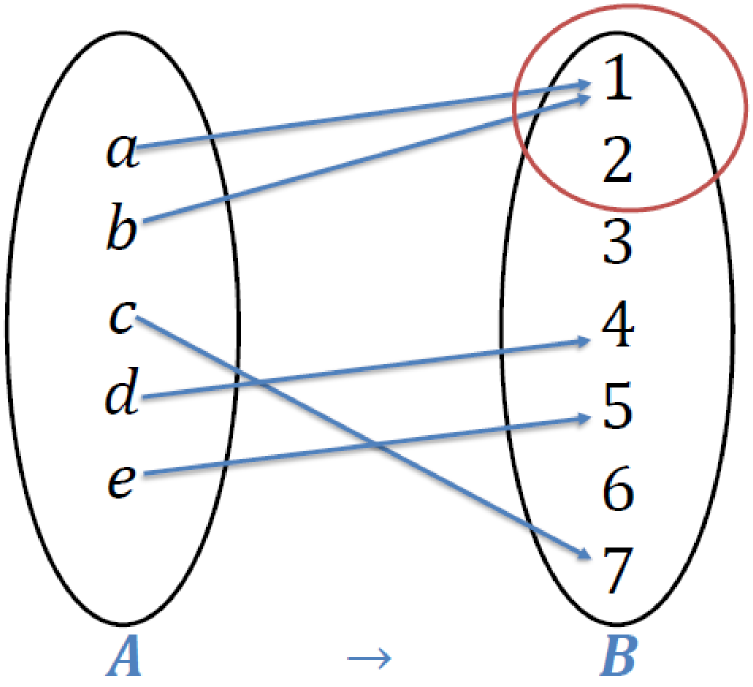
$f(b) = 3$

$f(c) = 7$

$f(d) = 4$

$f(e) = 5$

One-to-One function



$f(a) = 1$

$f(b) = 1$

$f(c) = 7$

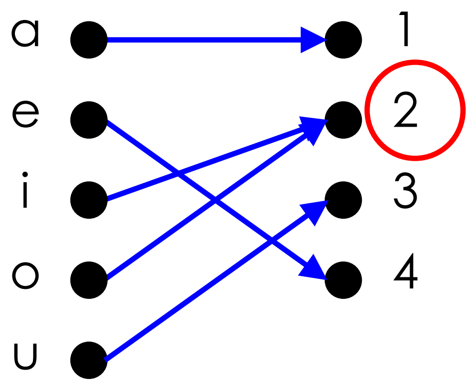
$f(d) = 4$

$f(e) = 5$

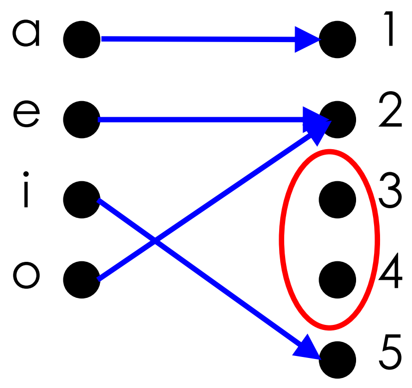
NOT One-to-One function

Onto Functions

- A function f from Ato B is called onto, or surjective, *if and only* if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$ [**The range equals to the codomain]**
- The elements [images] of Set B [Codomain] may have more than one pre-image from Set A [Domain]

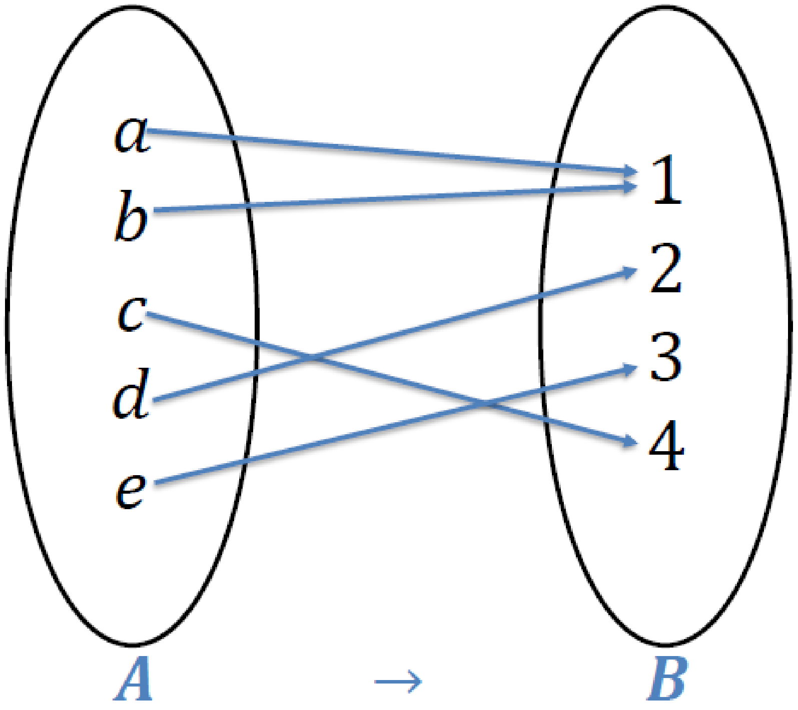


An onto function



A function that is not onto

More on Onto



$f(a) = 1$

$f(b) = 1$

$f(c) = 4$

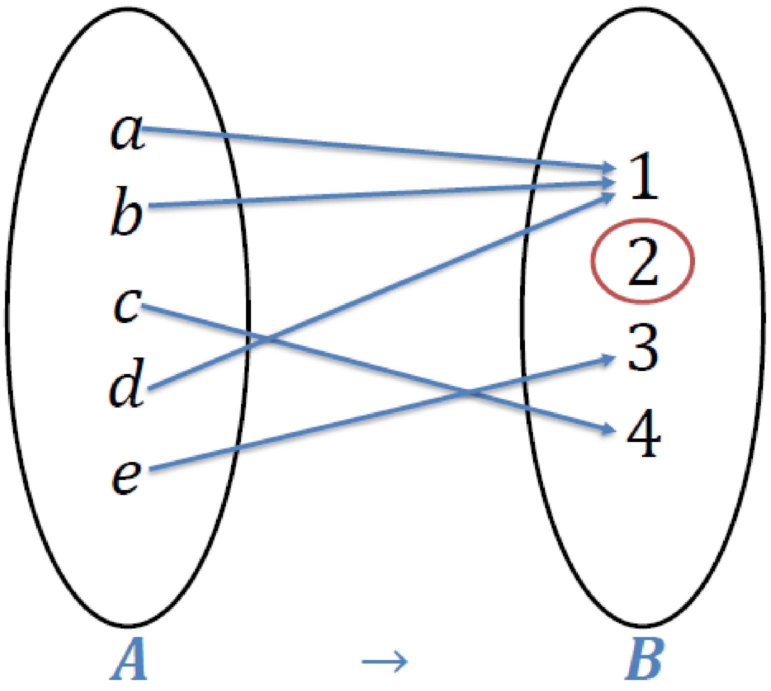
$f(d) = 2$

$f(e) = 3$

Co-Domain = {1,2,3,4}

Range = {1,2,3,4}

Onto function



$f(a) = 1$

$f(b) = 1$

$f(c) = 4$

$f(d) = 1$

$f(e) = 3$

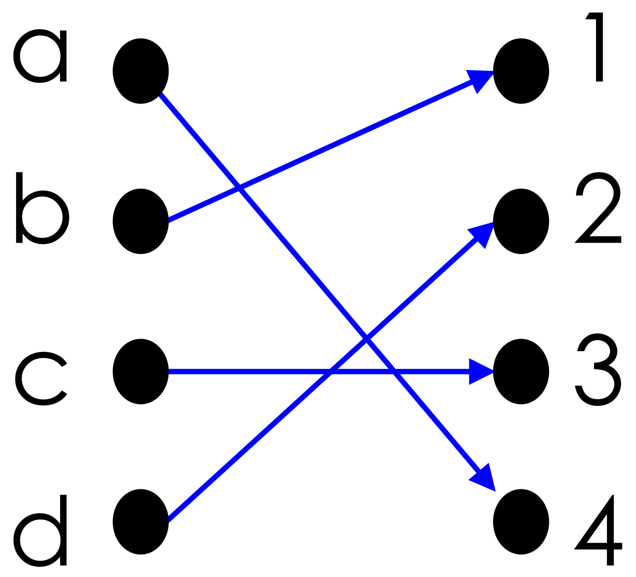
Co-Domain = {1,2,3,4}

Range = {1,3,4}

Not Onto function

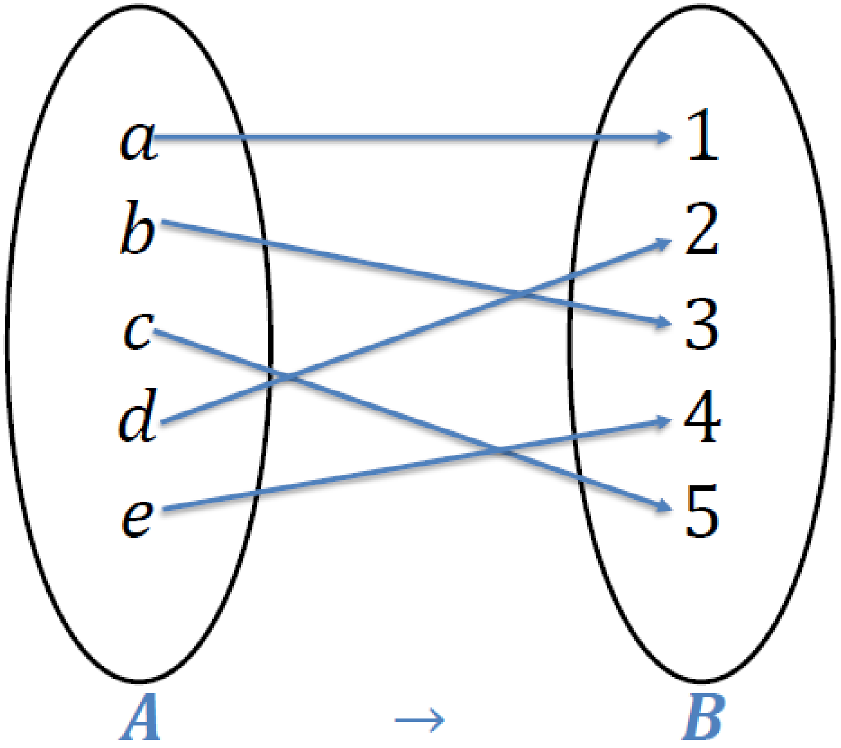
Bijections

- Consider a function that is **both one-to-one** and **onto**:



- Such a function is a **one-to-one correspondence**, or a **bijection**

Bijections



$f(a) = 1$

$f(b) = 3$

$f(c) = 5$

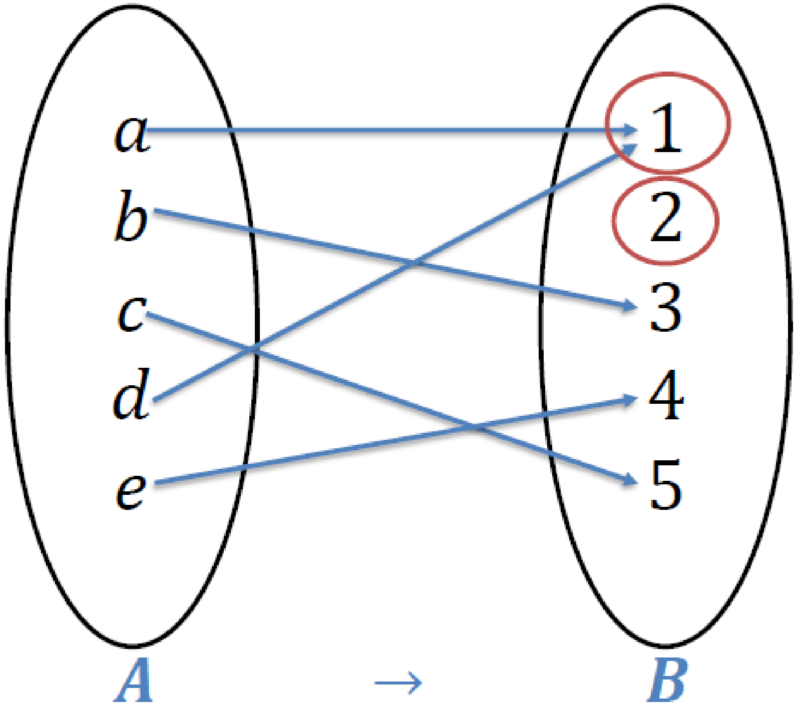
$f(d) = 2$

$f(e) = 4$

Co-Domain = {1,2,3,4,5}

Range = {1,2,3,4,5}

Bijection function



$f(a) = 1$

$f(b) = 3$

$f(c) = 5$

$f(d) = 1$

$f(e) = 4$

Co-Domain = {1,2,3,4,5}

Range = {1,3,4,5}

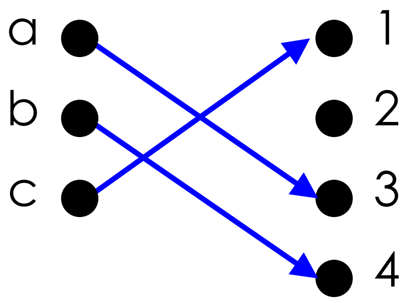
Not Bijection function

One-to-One, Onto, and Bijection

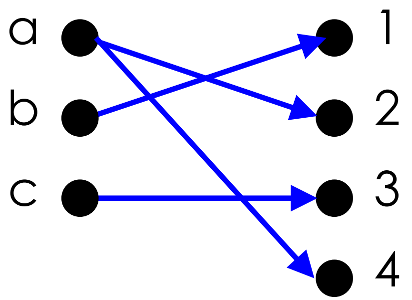
- **One to one**
  - Basic: All the images of Set B [Codomain] have a unique pre-image from Set A [domain]
  - Optional: There can be unused elements in Set B [Codomain]
- **Onto**
  - Basic: The range equals to the codomain
  - Optional: The images of Set B [Codomain] may have more than one pre-image from Set A [domain]
- **Bijection**
  - Both One to One [Basic] and Onto [Basic]

Onto vs. One-to-One

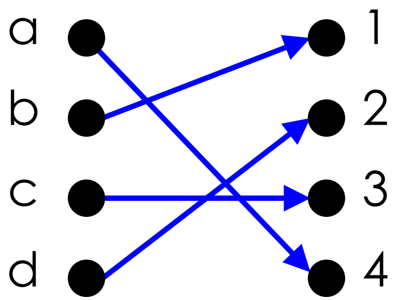
- Are the following functions onto, one-to-one, both, or neither?



1-to-1, not onto

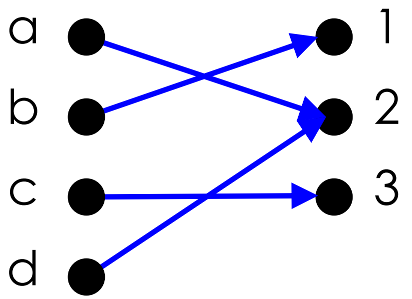


Not a valid function

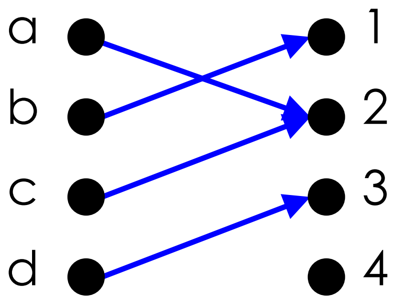


Both 1-to-1 and onto

Bijection



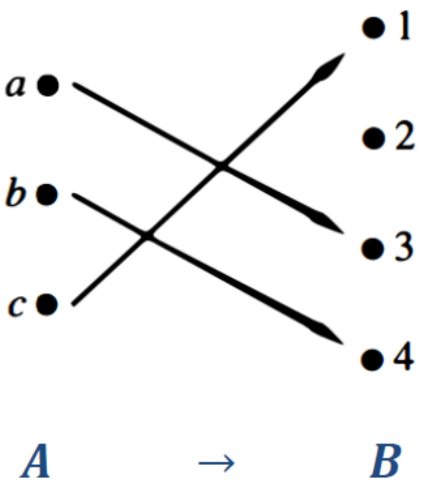
Onto, not 1-to-1



Neither 1-to-1 nor onto

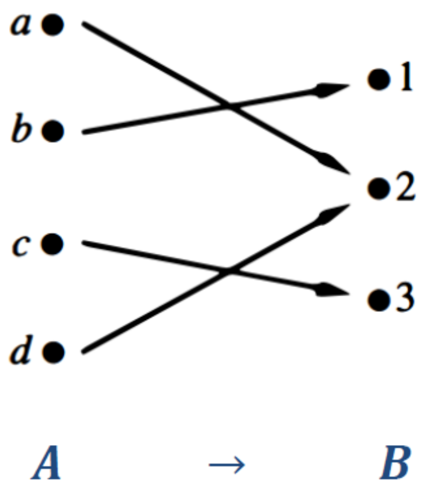
Onto vs. One-to-One

- Are the following functions onto, one-to-one, both, or neither?



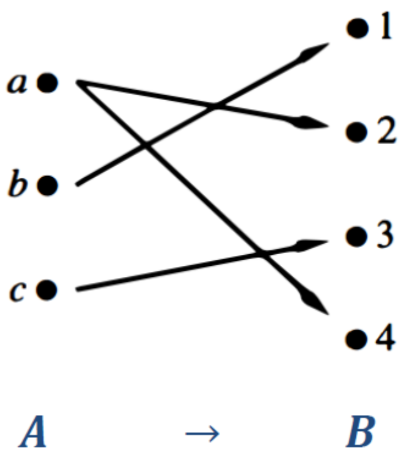
One-to-one

NOT onto

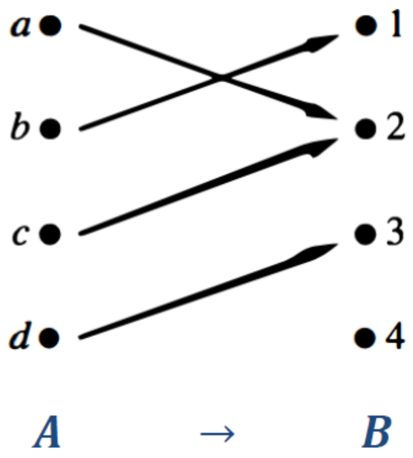


NOT One-to-one

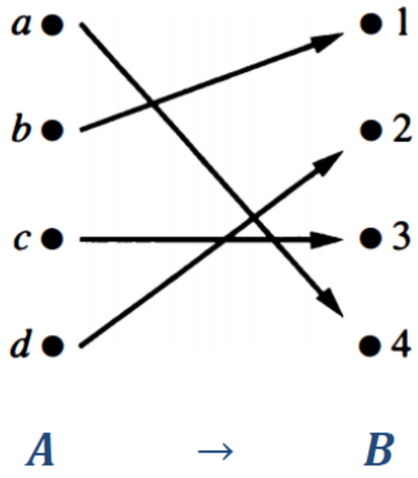
Onto



NOT a valid function



NOT One-to-one
NOT Onto



bijection

Example

- Determine whether the function $f(x) = x^2$ from the set of integers (Z) to the set of integers (Z) is one-to-one or onto.

Solution

- The function $f(x) = x^2$ is not one-to-one because the images of the co-domain do not have a unique pre-image. For instance, $f(1) = 1$ and $f(-1) = 1$, but $1 \neq -1$.
- The function $f(x) = x^2$ is not onto because the negative values of the co-domain are not selected. Therefore, the range does not equal to the co-domain.
- **Note** that the function $f(x) = x^2$ with its domain restricted to **Z**+ is one-to-one.

Example

- Determine whether the function $f(x) = x^2$ from the set of integers (**Z**+) to the set of integers (**Z**+) is one-to-one or onto.

Solution

- The function $f(x) = x^2$ is one-to-one because all the images of the co-domain have a unique pre-image.
- The function $f(x) = x^2$ is not onto because the range does not equal to the co-domain. For instance, the 3 value in the codomain is not selected.

Example

- Determine whether the function $f\ (x) = x^3$ from the set of integers (Z) to the set of integers (Z) is one-to-one or onto.

Solution

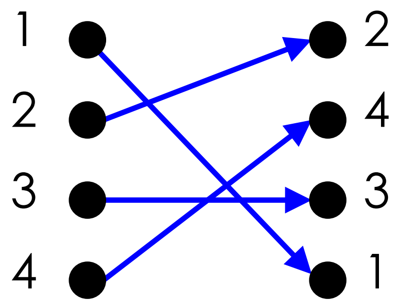
- The function $f\ (x) = x^3$ is one-to-one because all the images of the co-domain have a unique pre-image. For instance, f(1)=1 and f(-1)=-1

- The function $f\ (x) = x^3$ is not onto because the range does not equal to the co-domain. For instance, the 3 value in the codomain is not selected.

Identity functions

- A function such that the *image* and the *pre-image* are ALWAYS <u>equal</u>

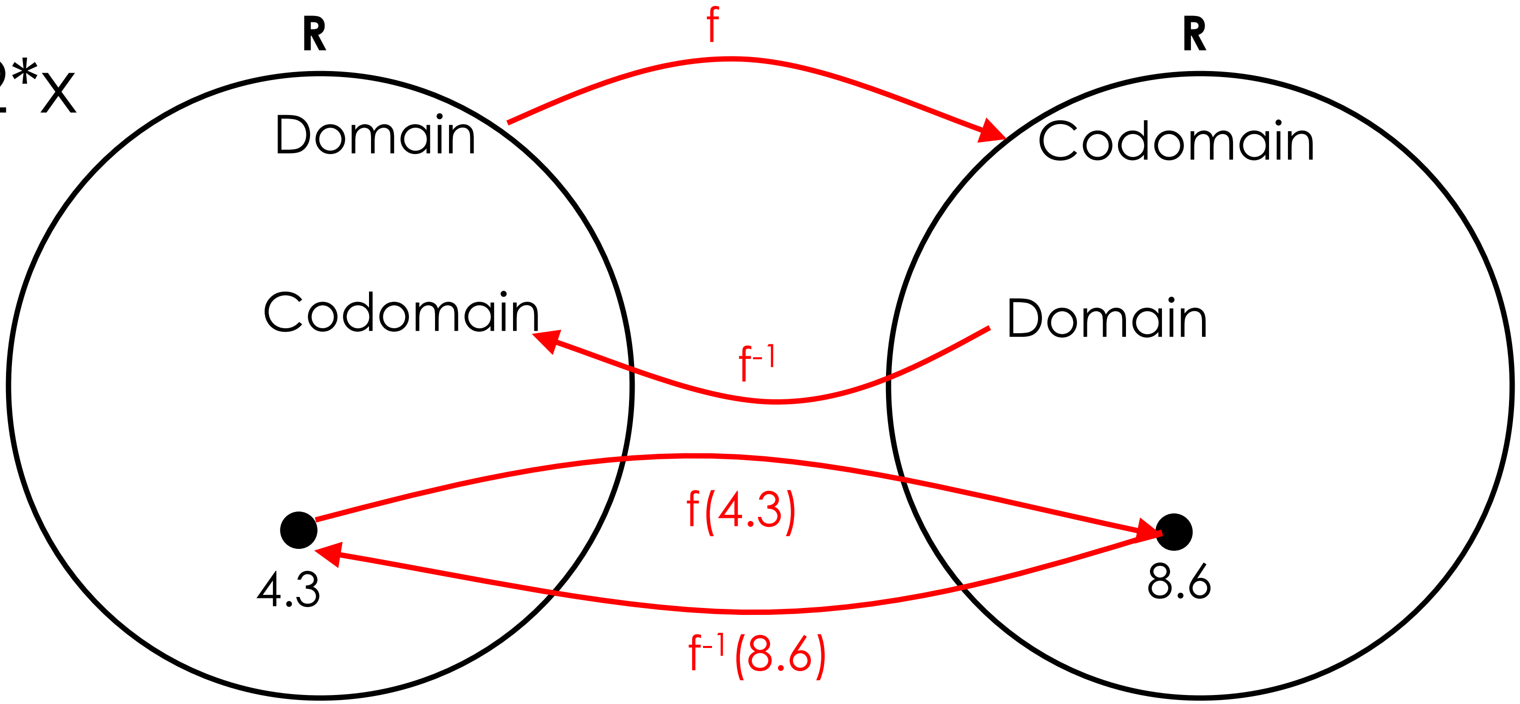
- f(x) = 1*x
- f(x) = x + 0



- The domain and the co-domain must be the same set
- *Note that the identity function is both one-to-one and onto (bijective).*

Inverse of a Function

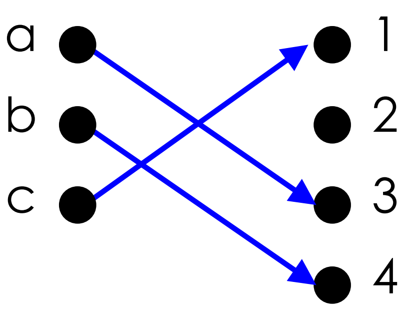
Let f(x) = 2*x



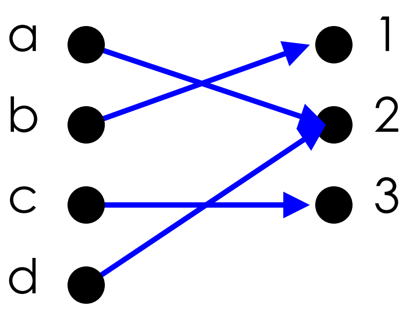
Then $f^{-1}(x) = x/2$

More on inverse functions

- Can we define the inverse of the following functions?



What is $f^{-1}(2)$?
Not onto!



What is $f^{-1}(2)$?
Not 1-to-1!

- An inverse function can ONLY be done defined on a bijection

Example

- Let $f: \mathbb{Z} \to \mathbb{Z}$ such that $f(x) = x+3$. Is f invertible? If so, what is the inverse?

- One-to-one?
  - $f(a)= f(b) \to a=b$
  - If $f(a) = 7$ and $f(b) = 7$, then $a+3=7$ and $b+3=7$
  - Thus $a=4$ and $b=4$
  - **All the images of the co-domain have a unique pre-image.**

Yes

Inverse

$x = f^{-1}(x) + 3$

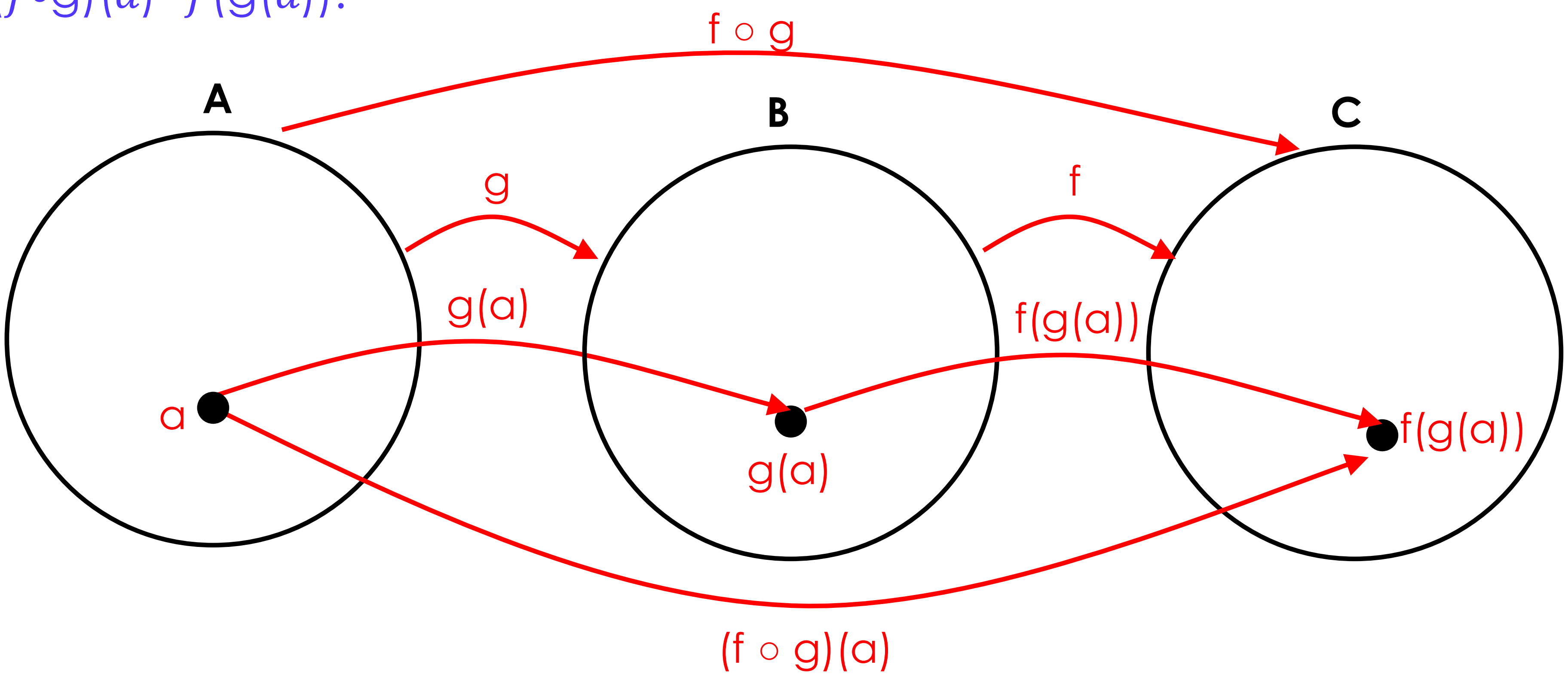
$f^{-1}(x) = x-3$

- Onto?
  - $4 \to 7$
  - $5 \to 8$
  - **The range equals to the codomain. All the values of the codomain are selected.**

Yes

Compositions of functions

- Let g be a function from the set A to the set B and let f be a function from the set B to the set C. The composition of the functions f and g, denoted by $f \circ g$, is defined by $(f \circ g)(a) = f(g(a))$.



Compositions of functions

- Let $(f \circ g)(x) = f(g(x))$
- Let $f(x) = 2x+3$ and $g(x) = 3x+2$.
- Find $(f \circ g)(1)$

- $g(1) = 3(1)+2 = 5$
- $f(g(1)) = f(5) = 2(5)+3 = 13$
- Thus, $(f \circ g)(1) = f(g(1)) = 13 \quad \equiv \quad f(g(x)) = 2(3x+2)+3 = 6x+7$

Compositions of functions

Does $f(g(x)) = g(f(x))$?

Let $f(x) = 2x+3$ Let $g(x) = 3x+2$

$f(g(x)) = 2(3x+2)+3 = \boxed{6x+7}$

$g(f(x)) = 3(2x+3)+2 = \boxed{6x+11}$

Not equal!

- Function composition is like Cartesian product not commutative! ($f \circ g \neq g \circ f$)

Example

If $f(x) = x+3$ and $g(x) = x^2-2$. find: $f \circ g(1)$ and $g \circ f(1)$

- $f \circ g(1) = f(g(x)) = f(1^2-2) = f(-1) = -1+3 = 2$

- $f \circ g(x) = f(x^2-2) = (x^2-2)+3 = x^2 + 1$
- $f \circ g(1) = 1^2 + 1 = 2$

- $g \circ f(1) = g(1+3) = g(4) = 4^2 - 2 = 14$

- $g \circ f(x) = g(x+3) = (x+3)^2 – 2 = x^2+6x+9-2 = x^2+6x+7$
- $g \circ f(1) = 1^2 +6(1)+7 = 1+6+7 = 14$

Some Useful functions

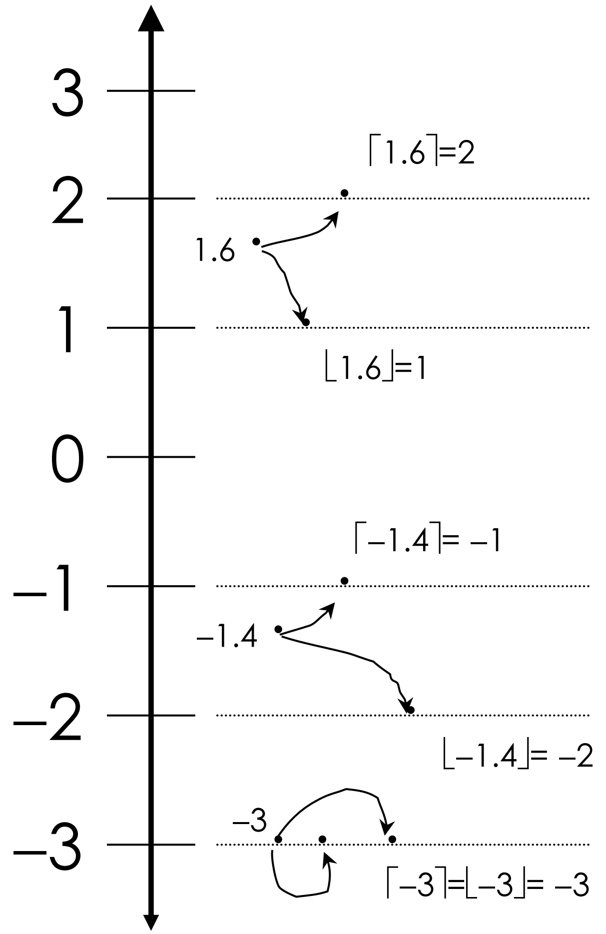
- Floor: $\lfloor x \rfloor$ round down the number ($x \notin Z$) to the nearest integer
- Ceiling: $\lceil x \rceil$ round up the number ($x \notin Z$) to the nearest integer

- Note that if $x \notin \mathbf{Z}$,

  $\lfloor -x \rfloor \neq -\lfloor x \rfloor$ && $\lceil -x \rceil \neq -\lceil x \rceil$

  $\lfloor -x \rfloor = -\lceil x \rceil$ && $\lceil -x \rceil = -\lfloor x \rfloor$

- Note that if $x \in \mathbf{Z}$,
  $\lfloor x \rfloor = \lceil x \rceil = x$.



Example

- Let f(x) = $\lfloor \frac{x^2}{2} \rfloor$. Find f(s) if S = {0, 1, 2, 3}
- f(0) = $\lfloor \frac{0^2}{2} \rfloor = \lfloor \frac{0}{2} \rfloor$ = 0
- f(1) = $\lfloor \frac{1^2}{2} \rfloor = \lfloor \frac{1}{2} \rfloor$ = 0
- f(2) = $\lfloor \frac{2^2}{2} \rfloor = \lfloor \frac{4}{2} \rfloor$ = 2
- f(3) = $\lfloor \frac{3^2}{2} \rfloor = \lfloor \frac{9}{2} \rfloor$ = 4

Example

$\lfloor 0.5 \rfloor =$

$\lceil 0.5 \rceil =$

$\lceil 3 \rceil =$

$\lfloor -0.5 \rfloor =$

$\lceil -1.2 \rceil =$

$\lfloor 1.1 \rfloor =$

$\lfloor 0.3 + 2 \rfloor =$

$\lceil 1.1 + \lceil 0.5 \rceil \rceil =$

$\lfloor 0.5 \rfloor = 0$

$\lceil 0.5 \rceil = 1$

$\lceil 3 \rceil = 3$

$\lfloor -0.5 \rfloor = -\lceil 0.5 \rceil = -1$

$\lceil -1.2 \rceil = -1$

$\lfloor 1.1 \rfloor = 1$

$\lfloor 0.3 + 2 \rfloor = 2$

$\lceil 1.1 + \lceil 0.5 \rceil \rceil = 3$

2.4 Sequence and Summation

- Sequences
- Arithmetic Progression
- Geometric Progression
- Special Integer Sequences
- Summations

Sequences

- A sequence represents an ordered list of elements.
- 1, 2, 3, 5, 8 is a sequence with five terms.
- 1, 3, 9, 27, 81 , ... , 3*n*, . . . is an infinite sequence.
- We use the notation $\{a_n\}$, to describe the sequence. $\{a_n\} = \{a_0 , a_1 , a_2 , a_3 , \ldots\}$ for n>=0,
  - where a_n is called the n^{th} term of the sequence.
  - $a_0, a_1, \ldots,$ represents the list of the terms of the sequence.
- Consider the sequence $\{a_n\}$, $a_n = n+5$, where $n \geq 0$. The first 5 terms of the sequence are {5, 6, 7, 8, 9 … } for n>=0

Arithmetic Progressions

- An arithmetic progression is a sequence of numbers such that the difference between the consecutive terms is constant.

  $$a,\ a+d,\ a+2d,\ a+3d,\ \ldots,\ a+nd,\ \ldots$$

  where:
  - $a \in \mathbb{R}$ is called the initial term
  - $d \in \mathbb{R}$ is called the common difference

- The general notation of the arithmetic progression is $\{a_n\} = a + nd$, where $n \geq 0$ (n is the index of the term).

Arithmetic Progressions: Examples

- Give the initial term and the common difference of
  - $\{s_n\}$ with $s_n = -1 + 4n$ -1 and 4 $-1, 3, 7, 11, \ldots,$
  - $\{t_n\}$ with $t_n = 7 - 3n$ 7 and -3 $7, 4, 1, -2, \ldots$

Geometric Progressions

- A geometric progression is a sequence of numbers where each term (after the first) is found by *multiplying the previous one by a fixed*, non-one *number* called the *common ratio*.

$$a, ar, ar^2, ar^3, \ldots, ar^n, \ldots$$

  where:
  - a is called the initial term
  - r is called the common ratio

- The general notation of the geometric progression is

  $\{a_n\} = a\,r^n$, where $n \geq 0$ (n is the index of the term).

Geometric Progressions: Examples

- Give the initial term and the common ratio of
  - $\{b_n\}$ with $b_n = (-1)^n$ — 1 and −1 — $1, -1, 1, -1, 1, \ldots$
  - $\{c_n\}$ with $c_n = 2(5)^n$ — 2 and 5 — $2, 10, 50, 250, 1250, \ldots$
  - $\{d_n\}$ with $d_n = 6(1/3)^n$ — 6 and 1/3 — $6, 2, 2/3, 2/9, 2/27, \ldots$

Recurrence Relations

- A *recurrence relation* for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \ldots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer.
- A sequence is called a *solution* of a recurrence relation if its terms satisfy the recurrence relation.

Fibonacci sequence

- Sequences can be neither geometric nor arithmetic
  - Sequence: { 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, … }
  - $F_n = F_{n-1} + F_{n-2}$, where $n \geq 2$
  - The first two terms (F_0 and F_1) are 1
  - Each term is the sum of the previous two terms

Example 1

- Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 1, 2, 3, \ldots,$ and suppose that $a_0 = 2$.
- What are a_1, a_2, and a_3?

Solution

We see from the recurrence relation that

- $a_1 = a_0 + 3 = 2 + 3 = 5$
- $a_2 = 5 + 3 = 8$
- $a_3 = 8 + 3 = 11.$

Example 2

- Let $\{a_n\}$ be a sequence that satisfies the recurrence relation

$a_n = a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \ldots,$

and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3?

Solution:

- $a_2 = a_1 - a_0 = 5 - 3 = 2$
- $a_3 = a_2 - a_1 = 2 - 5 = -3.$
- We can find a_4, a_5, and each successive term in a similar way.

Useful sequences

- $n^2 = 1, 4, 9, 16, 25, 36, \ldots$
- $n^3 = 1, 8, 27, 64, 125, 216, \ldots$
- $n^4 = 1, 16, 81, 256, 625, 1296, \ldots$
- $2^n = 2, 4, 8, 16, 32, 64, \ldots$
- $3^n = 3, 9, 27, 81, 243, 729, \ldots$
- $n! = 1, 2, 6, 24, 120, 720, \ldots$

Determining the sequence formula

- **Given values in a sequence, how do you determine the formula?**
- **Steps to consider:**
  - Is it an arithmetic progression (each term a constant amount from the last)?
  - Is it a geometric progression (each term a factor of the previous term)?
  - Does the sequence repeat itself (or cycle)?
  - Does the sequence combine previous terms?
  - Are there runs of the same value?

Determining the sequence formula

a) 3, 6, 12, 24, 48, 96, 192, …
- Each term is twice the previous: geometric progression
- $a_n = 3*2^n$, *where* $n \geq 0$ *Or* $a_n = 3*2^{n-1}$, *where* $n \geq 1$

b) 15, 8, 1, -6, -13, -20, -27, …
- Each term is 7 less than the previous term
- a_n = 15 – 7n, where n ≥ 0 Or a_n = 22 – 7n, where n ≥ 1

c) 1, 0, 2, 0, 4, 0, 8, 0, 16, 0, …
- The non-0 numbers are a geometric sequence (2^n) interspersed with zeros

d) 1, 2, 2, 3, 4, 4, 5, 6, 6, 7, 8, 8, …
- This sequence increases by one but repeats all even numbers once

Determining the sequence formula

e) 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, …

- The sequence alternates 1's and 0's, increasing the number of 1's and 0's each time

f) 2, 16, 54, 128, 250, 432, 686, …

- Each term is twice the cube of *n, where n is the index of the term*
- $a_n = 2*n^3$, where $n \geq 1$

g) 2, 3, 7, 25, 121, 721, 5041, 40321

- Each successive term is about factorial of n added by 1
- $a_n = n! + 1$, where $n \geq 1$

Summations

upper limit

- A summation:

$$\sum_{j=m}^{n} a_j \quad \text{or} \quad \sum\nolimits_{j=m}^{n} a_j$$

index of summation

lower limit

- is like a for loop:

```
int sum = 0;
for ( int j = m; j <= n; j++ )
    sum += a(j);
```

Evaluating sequences

$$\sum_{k=1}^{5}(k+1)$$

- 2 + 3 + 4 + 5 + 6 = 20

$$\sum_{j=0}^{4}(-2)^j$$

- $(-2)^0 + (-2)^1 + (-2)^2 + (-2)^3 + (-2)^4 = 11$

$$\sum_{i=1}^{10}3$$

- 3 + 3 + 3 + 3 + 3 + 3 + 3 + 3 + 3 + 3 = 30

$$\sum_{j=0}^{8}\left(2^{j+1}-2^j\right)$$

- $(2^1-2^0) + (2^2-2^1) + (2^3-2^2) + \dots (2^{10}-2^9) = 511$

Evaluating sequences

Given S = { 1, 3, 5, 7 }, evaluate the following summations:

- $\Sigma_{j \in S}\, j$
  - $1 + 3 + 5 + 7 = 16$
- $\Sigma_{j \in S}\, j^2$
  - $1^2 + 3^2 + 5^2 + 7^2 = 84$
- $\Sigma_{j \in S}\, (1/j)$
  - $1/1 + 1/3 + 1/5 + 1/7 = 176/105$
- $\Sigma_{j \in S}\, 1$
  - $1 + 1 + 1 + 1 = 4$

Summation of a geometric series

- Sometimes we can express a summation in closed form, as for geometric series
- Sum of a geometric series:

$$\sum_{j=0}^{n} ar^{j} = \begin{cases} \dfrac{ar^{n+1}-a}{r-1} & \text{if } r \neq 1 \\ (n+1)a & \text{if } r = 1 \end{cases}$$

- Example:

$$\sum_{j=0}^{10} 2^{n} = \frac{2^{10+1}-1}{2-1} = \frac{2048-1}{1} = 2047$$

More Series

$$\sum_{k=0}^{n} ar^k = a(r^{n+1}-1)/(r-1), r \neq 1$$

Geometric series

$$\sum_{k=1}^{n} k = n(n+1)/2$$

Arithmetic Series

(Euler's trick)

$$\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6$$

Quadratic series

$$\sum_{k=1}^{n} k^3 = n^2(n+1)^2/4$$

Cubic series

Review Questions

Problem 1

1. Is the function $f : \{a, b, c, d\} \rightarrow \{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$, and $f(d) = 3$ *one-to-one/Onto*? !

One to One because all the images of the co-domain have a unique pre-image.

Not Onto because the range does not equal to the codomain. The 2 value in the codomain is not selected.

Problem 2

- let f be a function from X={0, 1, 2, 3} to Y = {a, b, c} defined by f(0)=c, f(1)=b, f(2)=b and f(3)=c.
- Is f: X → Y either one-to-one or onto?

Not One to One because the images of the co-domain do not have a unique pre-image. For instance, f(0)=c and f(3)=c.

Not Onto because the range does not equal to the codomain. The a value in the codomain is not selected.

Problem 3

- Let $f : Z \rightarrow Z$ such that $f(x) = x^2$ from the set of integers to the set of integers. Is f one-to-one or onto?

- One-to-one => $f(a) = f(b)$ then $a = b$
  - If a=2 and b=-2
  - $f(a) = f(b)$ while $a \neq b$
  - The images of the co-domain do not have a unique pre-image. For instance, $f(2)=f(-2)=4$

Not One-to-One

- Onto
  - $f(a) \neq -1$.
  - The negative values of the co-domain are not selected. Therefore, the range does not equal to the co-domain.

Not Onto

Problem 4

- Determine whether each of these functions from {a, b, c, d} to itself is one-to-one or onto.
  a) f(a) = b, f(b) = a, f(c) = c, f(d) = d
  b) f(a) = b, f(b) = b, f(c) = d, f(d) = c
  c) f(a) = d, f(b) = b, f(c) = c, f(d) = d

a) One to One & Onto
b) Not One to One & Not Onto
c) Not One to One & Not Onto

- Determine whether each of these functions from $\mathbb{Z}$ to $\mathbb{Z}$ is one-to-one or onto.
  a) $f(n) = n + 1$ b) $f(n) = n^2 + 1$
  c) $f(n) = n^3$ d) $f(n) = \lceil n/2 \rceil$

a) One to One & Onto
b) Not One to One & Not Onto
c) One to One & Not Onto
d) Not One to One & Onto

Problem 5

- Find $f \circ g$ and $g \circ f$, where $f(x)=x^2 + 1$ and $g(x)=x + 2$, are functions from $\mathbb{R}$ to $\mathbb{R}$.
- $f \circ g = f(g(x)) = f(x+2) = (x+2)^2 + 1 = x^2 + 4 + 2x + 1 = x^2 + 2x + 5$
- $g \circ f = g(f(x)) = g(x^2 + 1) = x^2 + 1 + 2 = x^2 + 3$

Problem 6: Find these values

- $\lfloor 1.1 \rfloor$ — 1
- $\lceil 1.1 \rceil$ — 2
- $\lfloor -0.1 \rfloor$ — -1
- $\lceil -0.1 \rceil$ — 0
- $\lceil 2.99 \rceil$ — 3
- $\lceil -2.99 \rceil$ — -2
- $\lfloor \frac{1}{2} + \lceil \frac{1}{2} \rceil \rfloor$ — $\lfloor \frac{1}{2} + 1 \rfloor = \lfloor 3/2 \rfloor = 1$
- $\lceil \lfloor \frac{1}{2} \rfloor + \lceil \frac{1}{2} \rceil + \frac{1}{2} \rceil$ — $\lceil 0 + 1 + \frac{1}{2} \rceil = \lceil 3/2 \rceil = 2$

Supplementary Material

- Discrete Math - 2.3.1 Introduction to Functions
- Discrete Math - 2.3.2 One to One and Onto Functions
- Discrete Math - 2.3.3 Inverse Functions and Compositions Functions
- Discrete Math - 2.3.4 Useful Functions to Know

Supplementary Material

- Discrete Math - 2.4.1 Introduction to Sequences
- Discrete Math - 2.4.2 Recurrence Relations
- Discrete Math - 2.4.3 Summations and Sigma Notation
- Discrete Math - 2.4.4 Summation Properties and Formulas

See you next lecture